

Question 1. (15 Marks) [START A NEW PAGE]	Marks
a) Given that $z = 3 - 4i$ and $w = 1 + 2i$, write down the values of each of the following, expressing your answers in simplest form.	5
i) $\bar{z}$ ii) z^2 iii) $ z $ iv) $\frac{z}{w}$ v) $\arg(w)$	
b) (Detach and use the sheet provided) On the Argand diagram provided, the points A and B represent the complex numbers α and β respectively. On the same sheet construct, with the aid of geometrical instruments, points P, Q, R, S and T which respectively represent:	5
i) $\alpha + \beta$ ii) $-\beta$ iii) $\alpha - \beta$ iv) $2i\beta$ v) $\bar{\alpha}$	
Clearly indicate any significant lengths and angles.	
(Ensure that the Argand diagram is handed in with the rest of question 1.)	
c) i) Find all real numbers x and y that satisfy $(x + iy)^2 = -3 + 4i$.	3
ii) Hence solve the equation $z^2 - 3z + (3 - i) = 0$	2
Question 2. (15 Marks) [START A NEW PAGE]	Marks
a) A complex root of $z^3 + z^2 + z - 39 = 0$ is known to be $z = 3i - 2$. Deduce the values of the other two roots, giving reasons for your answers.	3
b) i) If $z = (\cos \theta + i \sin \theta)$, use de Moivre's Theorem to prove that	3
$\left(z^n + \frac{1}{z^n} \right) = 2 \cos n\theta.$	
ii) Hence, or otherwise, prove that $8 \cos^4 \theta = \cos 4\theta + 4 \cos 2\theta + 3$.	2
c) z satisfies $ z - 4 + 2i = 3$ and w satisfies $\arg(w + 1 - i) = \frac{\pi}{4}$. The point P represents z on an Argand diagram and Q represents w .	
i) Sketch the loci of P and Q on an Argand diagram.	4
ii) If $w = x + iy$, write down the equation of the locus of Q in terms of x and y .	2
iii) Hence deduce the minimum value of $ z - w $.	1

Question 3. (15 Marks) [START A NEW PAGE]			Marks
a)	i)	Express $z = \frac{\sqrt{3}+i}{1+i}$ in mod-arg form.	3
	ii)	Hence find the smallest positive integer m for which z^m is real.	2
b)	i)	On an Argand diagram, sketch the locus of points which satisfy the relationship $ z-2 = z+1-3i $, clearly showing its position and orientation.	2
	ii)	On the same Argand diagram, sketch the locus of $z + \bar{z} = 4$.	2
	iii)	Shade the area where $ z-2 \leq z+1-3i $ and $z + \bar{z} \geq 4$ simultaneously.	1
c)	The points A, B and C represent z_1 , z_2 and $z_1 + z_2$ in an Argand diagram where z_1 and z_2 satisfy the relationship $\frac{(z_2 - z_1)}{(z_2 + z_1)} = ki$ where k is a positive number (and $i = \sqrt{-1}$).		
	i)	Illustrate such an arrangement on an Argand diagram, denoting clearly properties of the quadrilateral $OACB$ and its diagonals (O is the origin).	2
	ii)	What is the shape of the quadrilateral $OACB$ if $\alpha) k=1$ $\beta) k \neq 1$.	2
	iii)	Write down the area of the quadrilateral $OACB$ in terms of z_1 and z_2 .	1

Question 4. (15 Marks) [START A NEW PAGE]			Marks
a)	i)	Solve the equation $z^6 + 8 = 0$, writing the roots in mod-arg form.	3
	ii)	Illustrate these roots on a clear Argand diagram.	2
	iv)	Write $z^6 + 8$ in factorized form over the field of Real numbers.	3
b)	The diagram shows an isosceles triangle ABP in the Argand diagram, with base AB and $\angle APB = \alpha$. PM is the perpendicular bisector of AB and so bisects $\angle APB$. Given that A and B represent the complex numbers z_1 and z_2 respectively, find, in terms of z_1 , z_2 and α , the complex numbers represented by:		
	i)	the vector AM .	2
	ii)	the vector MP .	3

Hence show that P represents the complex number

$$\frac{1}{2}(1 - i \cot \frac{\alpha}{2})z_1 + \frac{1}{2}(1 + i \cot \frac{\alpha}{2})z_2 . \quad 2$$

(END OF EXAMINATION)

Student Number : _____

Use this diagram for **Question 1(b)** and hand it in with the rest of your answers to question 1.

TERM 4 2006 EXT 2

1 a) $z = 3 - 4i$ $w = 1 + 2i$

i) $\bar{z} = 3 + 4i$

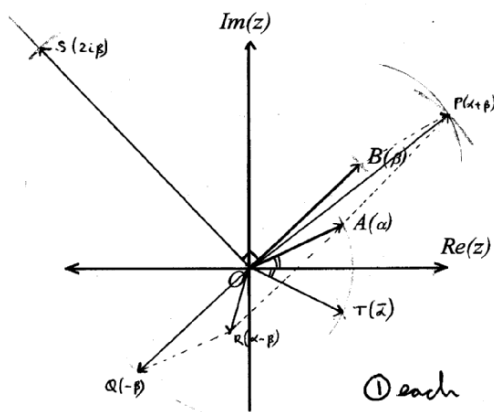
ii) $z^2 = 9 - 16 - 24i = -7 - 24i$

iii) $|z| = \sqrt{3^2 + (-4)^2} = 5$

iv) $\frac{z}{w} = \frac{(3-4i)(1-2i)}{(1+2i)(1-2i)}$ ① each
 $= \frac{3-8-4i-6i}{5} = \frac{-1-2i}{5}$

v) $\arg(w) = \tan^{-1}\left(\frac{2}{1}\right) = \tan^{-1} 2$

b)



c) $x^2 - y^2 = -3$ (Real)
 $2xy = 4$ (Imaginary)

$\therefore y = 2/x$

$x^2 - \frac{4}{x^2} = -3$

$x^4 + 3x^2 - 4 = 0$

$(x^2 - 1)(x^2 + 4) = 0$ ($x^2 \neq 4$)
 $\therefore x = \pm 1 \therefore y = \pm 2$

$(x, y) = (1, 2) \text{ or } (-1, -2)$

Quadratic formula

$z = \frac{3 \pm \sqrt{9 - 4(3-i)}}{2} = \frac{3 \pm \sqrt{-3+4i}}{2}$

$= \frac{3 \pm (1+2i)}{2} = 2+i \text{ or } 1-i$

2 a) As all coefficients are real, ①
 complex roots come in conjugate pairs

$\therefore$ Another root is $-2-3i$ ①

Sum of roots = $-\frac{b}{a} = -1$

$\therefore$ Third root is 3 ①

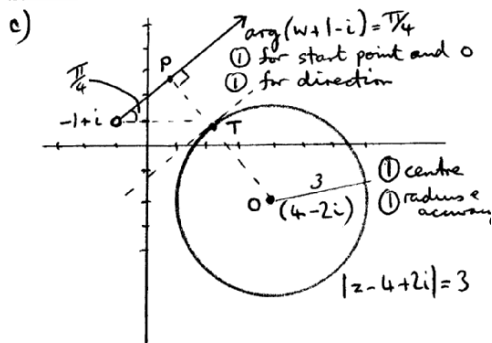
b) i) $z^n = (\cos \theta + i \sin \theta)^n$ +
 $= (\cos n\theta + i \sin n\theta)$ (de Moivre) ①
 $z^{-n} = (\cos(-n\theta) + i \sin(-n\theta))$ (de Moir)
 $= \cos n\theta - i \sin n\theta$ + ①

$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta$ (adding +) ①

ii) $\left(z + \frac{1}{z}\right)^4 = 16 \cos^4 \theta$

But $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$
 $= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$
 $= 2 \cos 4\theta + 8 \cos 2\theta + 6$

Combining & divide thro' by 2
 $8 \cos^4 \theta = \cos 4\theta + 4 \cos 2\theta + 3$



ii) gradient is 1, intercept 2 ①
 Eqn is $y = x + 2$, $x > -1$ ①

iii) $OP = \frac{|4 - (-2) + 2|}{\sqrt{1+1}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$

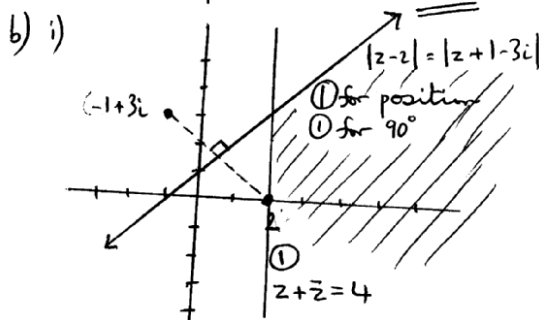
$\therefore PT = \min |z - w| = 4\sqrt{2} - 3$ ①

3 a) i) $z = \frac{2 \cos(\tan^{-1} \frac{1}{\sqrt{3}})}{\sqrt{2} \cos(\tan^{-1} 1)} = \frac{2 \cos(\frac{\pi}{6})}{\sqrt{2} \cos(\frac{\pi}{4})}$ ①
 $= \sqrt{2} \cos(\frac{\pi}{12})$ ①

ii) $z^m = (\sqrt{2})^m (\cos(\frac{m\pi}{12}) + i \sin(\frac{m\pi}{12}))$
 $\cos(\frac{m\pi}{12}) = \cos(\frac{m\pi}{12} + \pi) = \cos(\frac{m\pi}{12})$ ①

Which is real when $\sin(\frac{m\pi}{12}) = 0$ ii)

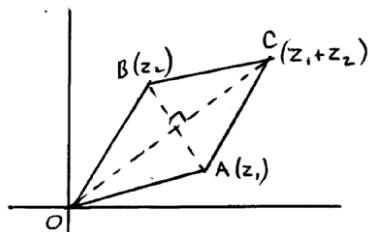
Smallest positive m is 12 ①



ii) $z + \bar{z} = 4 \Rightarrow 2 \operatorname{Re}(z) = 4 \Rightarrow \operatorname{Re}(z) = 2$ ①

iii) ① for correct area shaded

c) i) $z_2 - z_1$ and $z_2 + z_1$ are represented by the diagonals of the quadrilateral OACB. These diagonals are at 90° as their ratio is $k i$.



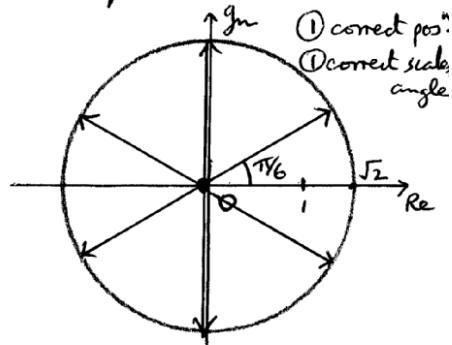
As $BC \parallel OA$ and $OB \parallel AC$, this is a parallelogram so $OB = AC$ and $OA = BC$. A parallelogram with diagonals at 90° ① is a rhombus. Thus $|z_1| = |z_2|$ ①

ii) a) $k = 1 \Rightarrow$ Square b) $k \neq 1 \Rightarrow$ Rhombus

iii) Area = $\frac{1}{2} |z_1 + z_2| |z_2 - z_1|$ ①

4 a) i) $z^6 = -8 = 8 \cos(\pi + 2k\pi)$ ① ($k \in \mathbb{Z}$)
 $\therefore z = \sqrt{2} \cos(\frac{\pi}{6} + \frac{k\pi}{3})$ ① (de Moivre)

$z = \sqrt{2} \cos(-\frac{\pi}{6}), \sqrt{2} \cos(-\frac{\pi}{2}), \sqrt{2} \cos(-\frac{\pi}{6})$
 $\sqrt{2} \cos(\frac{\pi}{6}), \sqrt{2} \cos(\frac{\pi}{2}), \sqrt{2} \cos(\frac{\pi}{6})$ ①
 (or equivalent)



iii) Pair off the linear roots

eg. $(z - \sqrt{2} \cos(-\frac{\pi}{6}))(z - \sqrt{2} \cos(\frac{\pi}{6}))$
 $= (z - \sqrt{2} \cos \frac{\pi}{6} + i\sqrt{2} \sin \frac{\pi}{6})(z - \sqrt{2} \cos \frac{\pi}{6} - i\sqrt{2} \sin \frac{\pi}{6})$
 $= (z^2 - 2\sqrt{2} \cos \frac{\pi}{6} z + 2) = (z^2 - z\sqrt{6} + 2)$

Similarly $\frac{\pi}{6} \rightarrow (z^2 + z\sqrt{6} + 2)$

Also $\frac{\pi}{2} \rightarrow (z^2 + 2)$ ①

$\therefore z^6 + 8 = (z^2 + 2)(z^2 + z\sqrt{6} + 2)(z^2 - z\sqrt{6} + 2)$

b) i) M represents $\frac{z_1 + z_2}{2}$ ①

$\therefore AM$ represents $\frac{z_1 + z_2}{2} - z_1$
 $= \frac{z_2 - z_1}{2}$ ①

ii) MP is AM rotated by 90° (a-c) and enlarged in ratio $\frac{|PM|}{|AM|} = \cot \alpha$

$\therefore MP$ represents $\frac{(z_2 - z_1)}{2} i \cot \frac{\alpha}{2}$ ①

P represents $OM + MP$ ①
 $= \frac{z_1 + z_2}{2} + \frac{(z_2 - z_1)}{2} i \cot \frac{\alpha}{2}$
 $= \frac{1}{2} z_1 (1 - i \cot \frac{\alpha}{2}) + \frac{1}{2} (1 + i \cot \frac{\alpha}{2}) z_2$